

Field and Galois Theory

Prelude

1.

- (a) Give an example of a tower of field extensions $F \supset L \supset K$ such that L/K is Galois and F/L is Galois, but F/K is not Galois.
- (b) Give an example of a primitive element in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$. Prove that the element you give is indeed a primitive element. Find the minimal polynomial of the element you find.

2. Let ζ be a primitive k th root of unity.

- (a) Prove that $K = \mathbb{Q}(\zeta)$ is a Galois extension of $\mathbb{Q}$.
- (b) Compute $\text{Gal}(K/\mathbb{Q})$.

Fugue

3. (January 2020 Problem 2)

- (a) Let a be a root of the polynomial $x^4 - 7 \in \mathbb{Q}[x]$. Show that $\mathbb{Q}(a)$ is not Galois over $\mathbb{Q}$.
- (b) Let F be a field of characteristic 5, and let b be a root of $x^4 - 7 \in F[x]$. Show that $F(b)$ is Galois over F .
- (c) Compute the Galois group $\text{Gal}(F(b)/F)$.

4. (January 2021 Problem 5) Let $\mathbb{F}_p$ denote the field with p elements.

- (a) What is the factorization of the polynomial $x^4 + 1$ into irreducible factors in $\mathbb{F}_3[x]$?
- (b) Give an example of an element $a \in \mathbb{F}_5$ such that $x^4 - a$ is irreducible in $\mathbb{F}_5[x]$, and prove that $x^4 - a$ is irreducible with this choice of a .
- (c) Prove that for any nonzero choice of $a \in \mathbb{F}_{103}$, the polynomial $x^4 - a$ factors either as a product of two irreducible quadratic polynomials, or as a product of two linear polynomials and an irreducible quadratic.

5. (January 2016 Problem 1) Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f .

- (a) Show that if $\text{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.
- (b) Is the converse statement true? That is, suppose $L = K(\alpha)$; must $\text{Gal}(L/K)$ be abelian?

6. (January 2017 Problem 4) This is a question about “biquadratic extensions” in two parts.

- (a) Let F/E be a degree-4 Galois extension, where E and F are fields of characteristic different from 2. Show that $\text{Gal}(F/E) \cong (\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$ if and only if there exist $x, y \in E$ such that $F = E(\sqrt{x}, \sqrt{y})$ and none of x, y, xy are squares in E .
- (b) Give an example of a field E of characteristic 2 that is not algebraically closed but that has no Galois extension F/E with Galois group $(\mathbb{Z}/2\mathbb{Z}) \times (\mathbb{Z}/2\mathbb{Z})$

7. Suppose that $K/\mathbb{Q}$ is a degree-4 Galois extension with Galois group $\mathbb{Z}/4\mathbb{Z}$. Prove that $\sqrt{-1} \notin K$.

8. (August 2018 Problem 4)

- (a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.
- (b) Let $K = \mathbb{F}_q(t)$, and let $L = \mathbb{F}_q(t^{1/p})$ where q is a power of p . Prove that there are no automorphisms of L fixing K .

9. (January 2016 Problem 5) Consider the group $\text{GL}_n(\mathbb{Q})$ of invertible $n \times n$ matrices with rational coefficients. Suppose $G \subset \text{GL}_n(\mathbb{Q})$ is a finite subgroup. Prove that every prime factor p of the order $|G|$ satisfies $p \leq n + 1$.